

Characteristics of patients who avoid wheat and/or gluten in the absence of Celiac disease.



BACKGROUND: Gastrointestinal symptoms that respond to the removal of wheat and/or gluten are becoming more common. Patients who avoid wheat and/or gluten (PWA WG) are a heterogeneous group and predominantly self-diagnosed prior to presenting for clinical evaluation. **SPECIFIC AIM:** We characterized PWA WGs seen at a tertiary care referral center and compared them to patients with celiac disease (CD) and subjects in the National Health and Nutrition examination survey (NHANES). **METHODS:** This was a cross-sectional study evaluating patients seen by four gastroenterologists at a CD referral center. Baseline characteristics, laboratory values, and medical comorbidities were compared to CD patients who presented at the same center and subjects enrolled in NHANES. **RESULTS:** Eighty-four PWA WGs were identified and compared to 585 CD patients and 2,686 NHANES patients. Thirty-two alternative diagnoses were made in 25 (30%) PWA WGs, including small intestinal bacterial overgrowth and fructose/lactose intolerance. When compared to patients with CD, PWA WGs had similar body mass index (BMI, 23.1 vs. 23.5, $p = 0.54$) and mean hemoglobin value (13.4 vs. 13.3, $p = 0.6$). When compared to male and female patients in NHANES, BMI, folate, and mean hemoglobin values were lower in PWA WGs. Both male and female PWA WGs had a lower prevalence of hypertension. **CONCLUSION:** While there are similarities between CD and PWA WGs that could possibly be due to shared HLA haplotypes or an effect of the gluten-free diet, alternative diagnoses are common in these patients. PWA WGs have a similar cardiovascular profile as CD patients in terms of lower BMI and lower prevalence of hypertension.